

Wafer Fault Detection using CNN with Explainable AI

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WAFER DATASET

Each input is a univariate time series representing the trace signal collected during a manufacturing step. The goal is to classify whether the trace is from a normal or abnormal wafer process.

Characteristic	Description
Samples	~1814 training/ ~6000+ testing
Input Shape	(152,1) – 152-time steps univariate signal
Each Sample	1D time series from a sensor monitoring semiconductor wafer processing
Tasks	Binary Classification: Normal vs Abnormal
Labels	Class 0 = Normal wafer trace Class 1= Abnormal wafer trace

WAFER DATASET

MODEL METRICS

Model Used : CNN
Model Accuracy - 0.61
Threshold - 0.5

Class	Precision	Recall	F1-Score
0 (Left)	0.97	0.99	0.98
1 (Right)	1	1	1

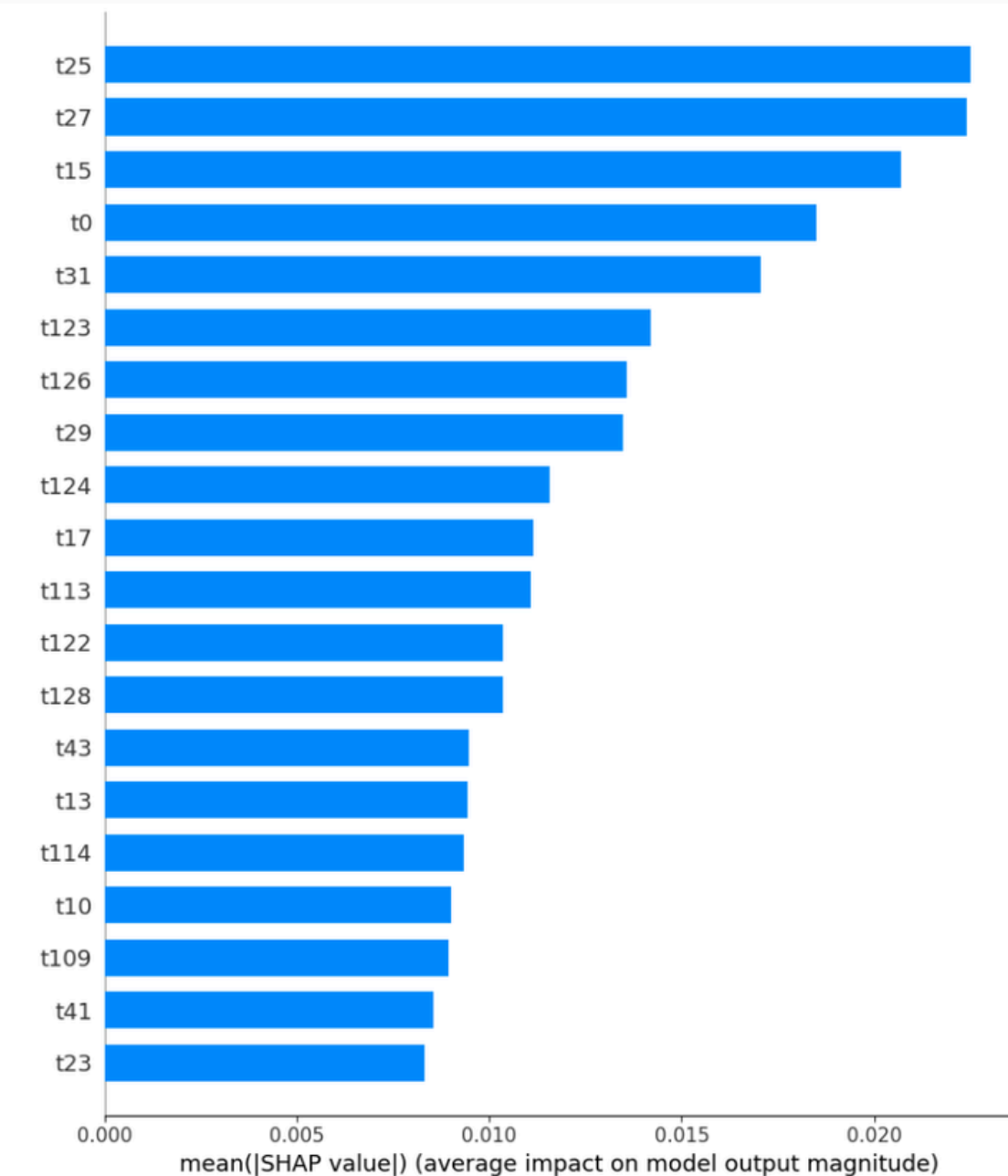
SHAP GLOBAL IMPORTANCE

TOP N FEATURES

This global explanation helps interpret which time windows the model relies on when distinguishing normal from abnormal wafer sensor traces—useful for quality control teams to understand and optimize key signal checkpoints.

The top-ranked feature is t25, closely followed by t27 and t15. This indicates that sensor signals collected around these specific time steps have the strongest influence on the model's decision boundary.

Early time steps like t0 and midpoints such as t31, t29, and t17 also appear in the top 10, suggesting that important discriminative features are not confined to a specific part of the signal window.



INTEGRATED GRADIENTS (Global Explanation)

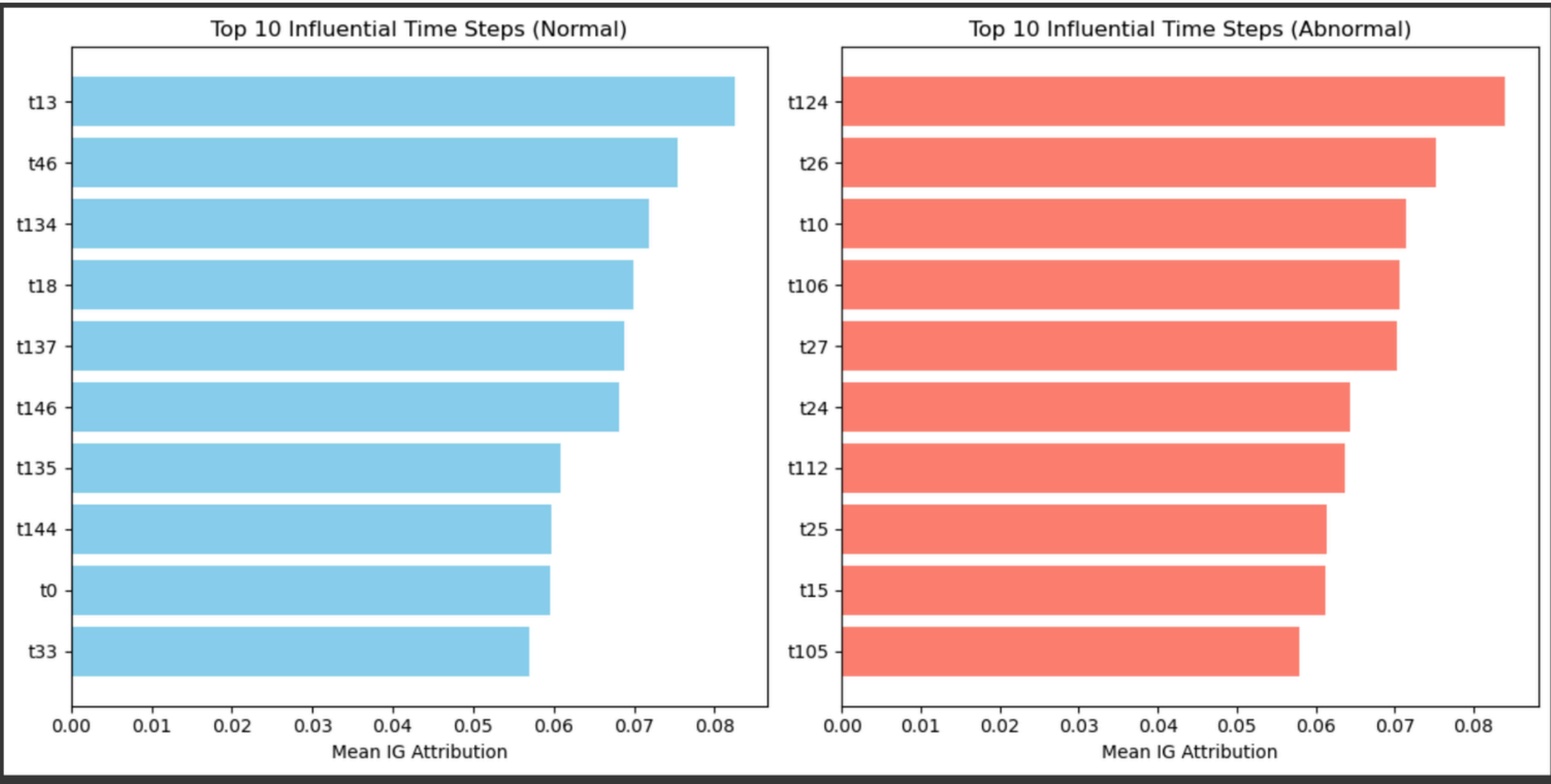
TOP N FEATURES BY CLASS

The global IG plot shows the average absolute attribution of each time step across all samples in the normal and abnormal classes of the Wafer dataset.

Top-K Influential Time Steps:

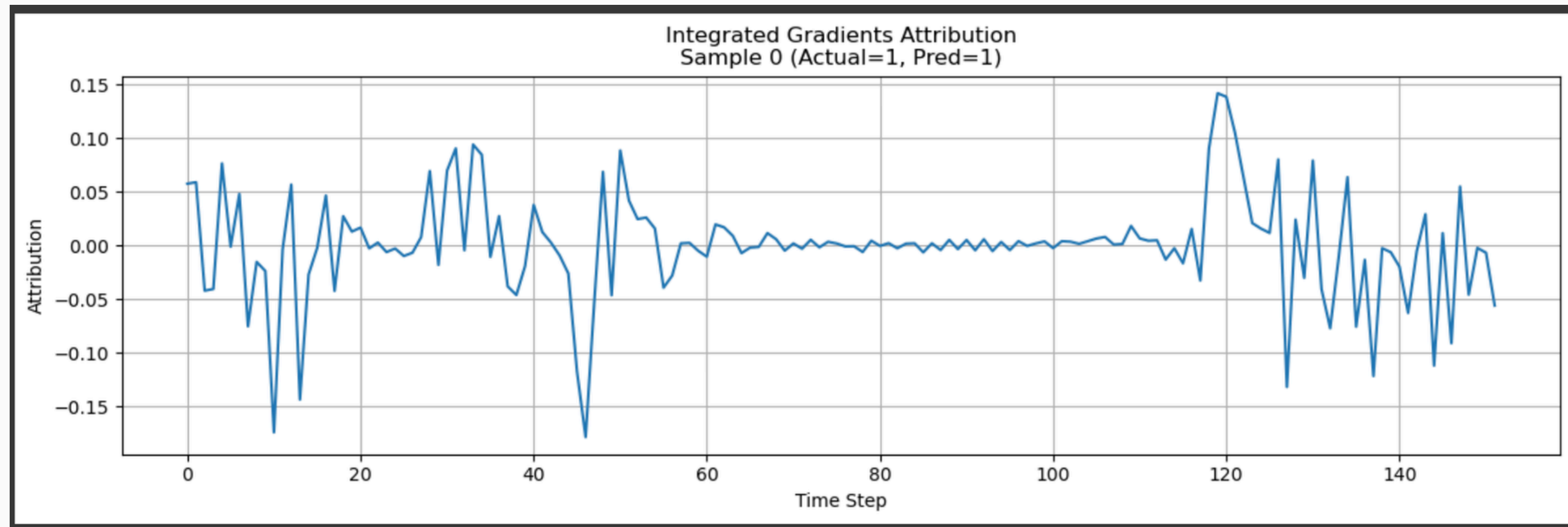
Normal: t13, t46, t134

Abnormal: t124, t26, t10



INTEGRATED GRADIENTS (Local Explanation)

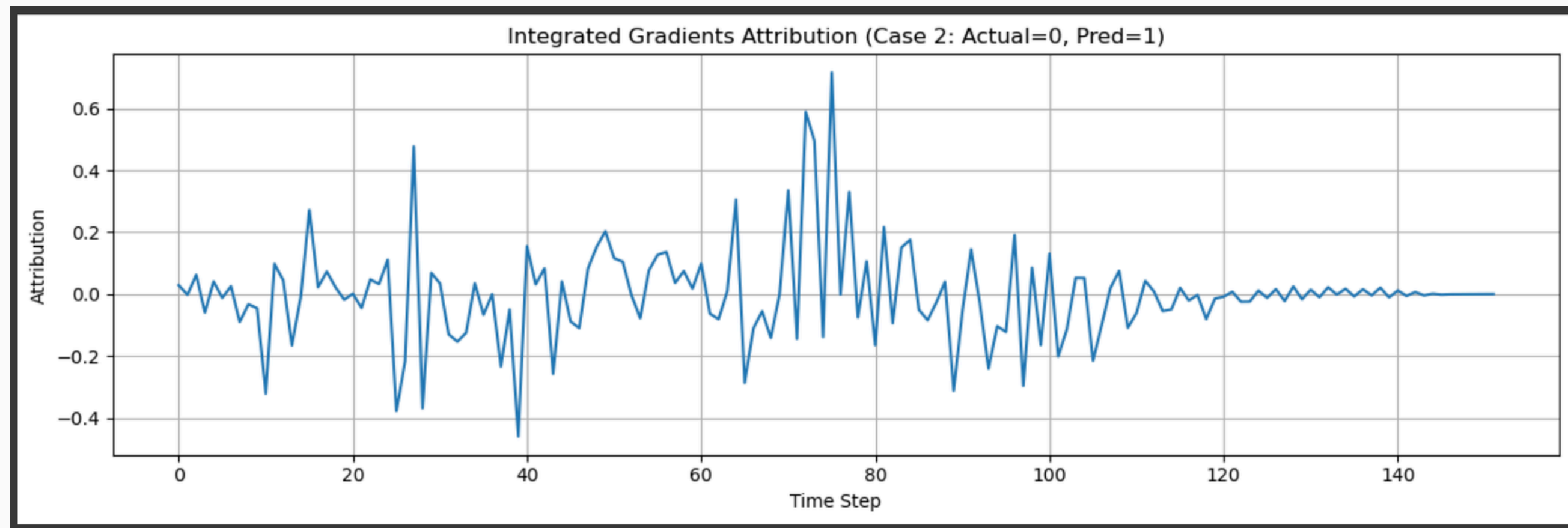
Example 1: (Actual = 1, Predicted = 1)



Interpretation: The model correctly identified this sample as class 1 by focusing heavily on the final third of the sequence, suggesting it relies on late temporal patterns to confirm the presence of an abnormal signal.

INTEGRATED GRADIENTS (Local Explanation)

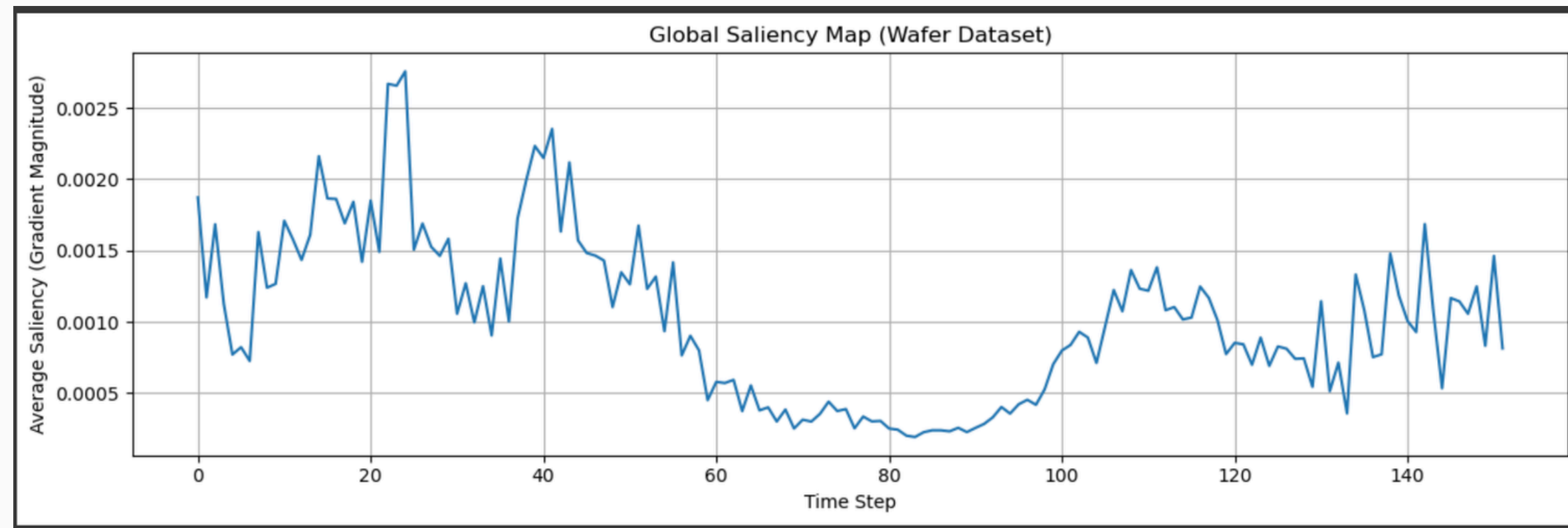
Example 2: (Actual = 0, Predicted = 1)



Interpretation: The model was misled by transient variations in the mid-sequence window and misclassified this as abnormal. The gradient attribution shows that false positives may stem from sensitivity to signal noise or spikes, rather than true underlying pattern deviations.

SALIENCY MAP (Global Explanation)

This saliency map **captures the average model sensitivity across time** by measuring the gradient magnitude with respect to input features.

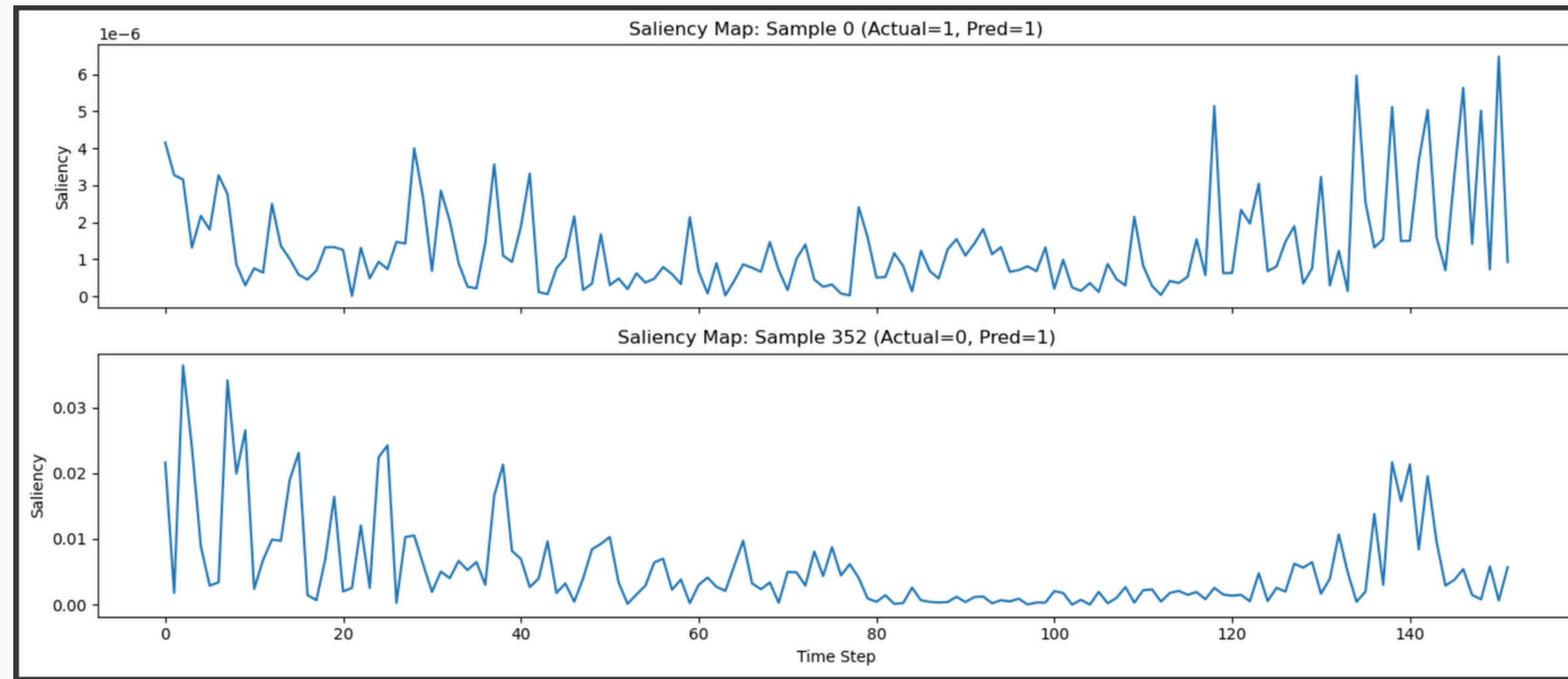


The model's attention is distributed, but with two key zones of activity:

- t20–t40: Early signal spikes.
- t120–t150: Final segment, often indicative of anomalies or steady deviation.

SALIENCY MAP (Local Explanation)

The local saliency maps demonstrate how the model focuses on different time steps when making individual predictions:



High and focused saliency at the end of the sequence reflects confident, correct decisions, while spread-out and noisy saliency indicates uncertainty and misclassification.

Conclusion

Wafer Dataset

- A CNN was applied to the Wafer dataset, achieving high accuracy and F1-score in classifying normal vs. abnormal signals.
- Integrated Gradients and Saliency Maps confirmed the model focused on meaningful signal regions, enhancing interpretability.
- The analysis provided insights into decision-making, emphasizing the value of explainable AI in critical domains like manufacturing.